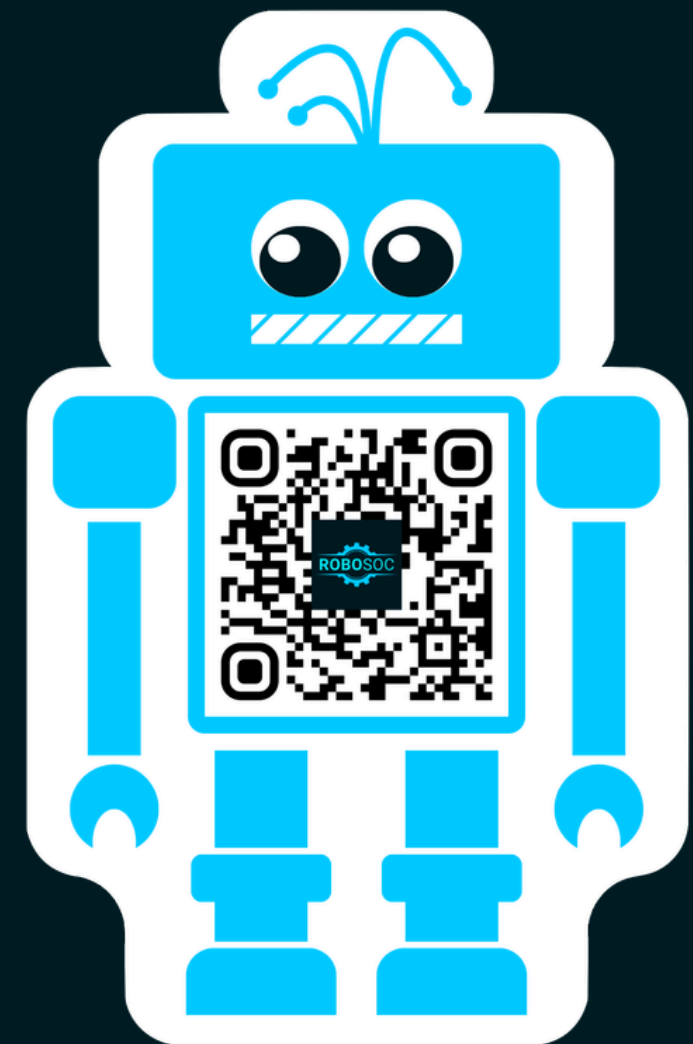


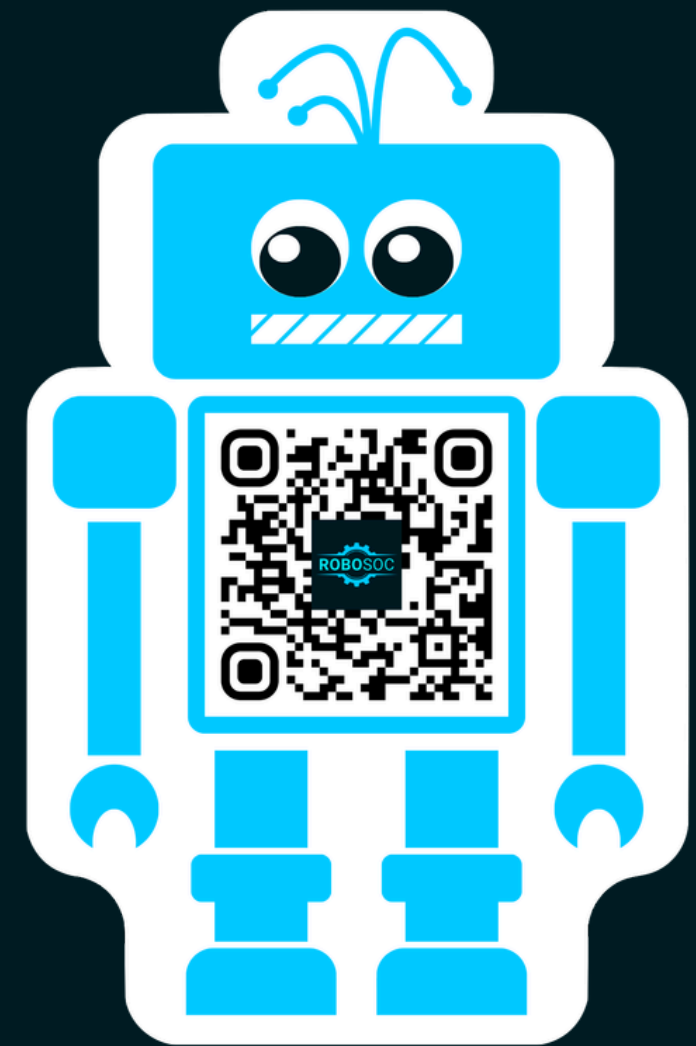


Competitions Talk





Workshops Overview





Workshop Programme

1. Arduino & ESP32

- Learn the basics of Arduino programming
- Connect inputs and outputs
- Control LEDs, sensors and other components

2. Circuit Diagrams

- Learn how to read and create circuit diagrams
- Build circuits using ESP32
- Circuit escape room & reaction-light challenge



Workshop Programme

3. Sensors

- Explore different types of sensors
- Build a smart night light
- Build a parking sensor
- Combine multiple sensors

4. Decisions & Automation

- Learn how robots make decisions
- Use sensors to respond to different situations
- Build a sensor-controlled crossing signal



Workshop Programme

5. Line Following & Navigation

- Learn different navigation techniques
- Experiment with ready-made robots
- Complete a line-following challenge

6. Motors

- Learn how servo and DC motors work
- Control motors using the ESP32
- Build a moving robot





Workshop Programme

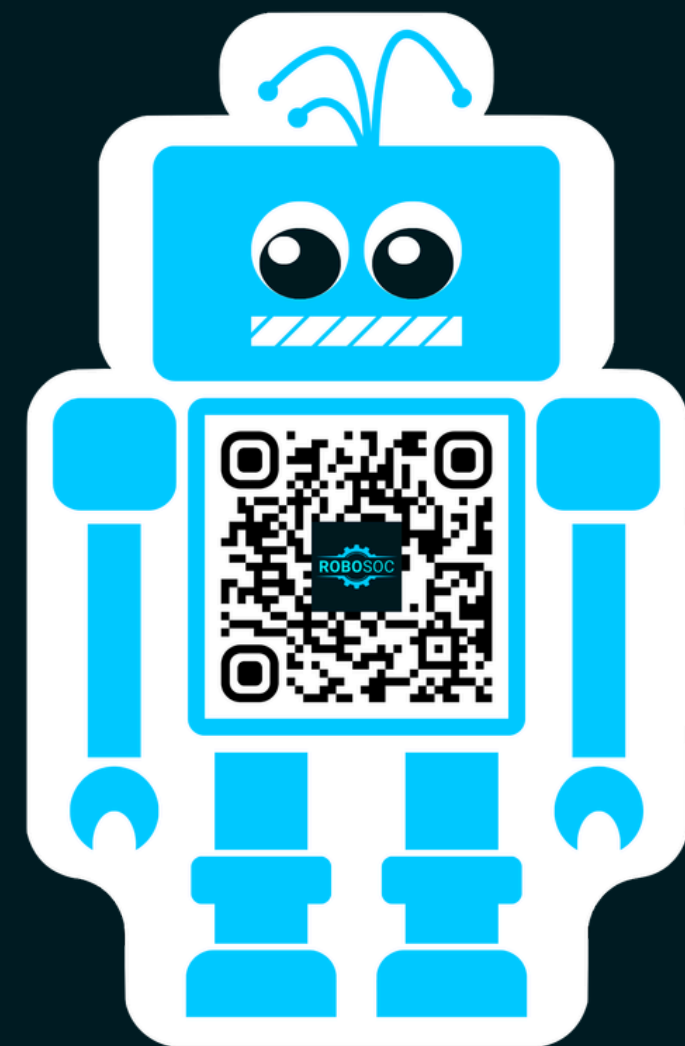
7. AI & Object Detection

- Introduction to AI vision
- Detect colours, shapes and objects
- Train and improve detection





Competitions



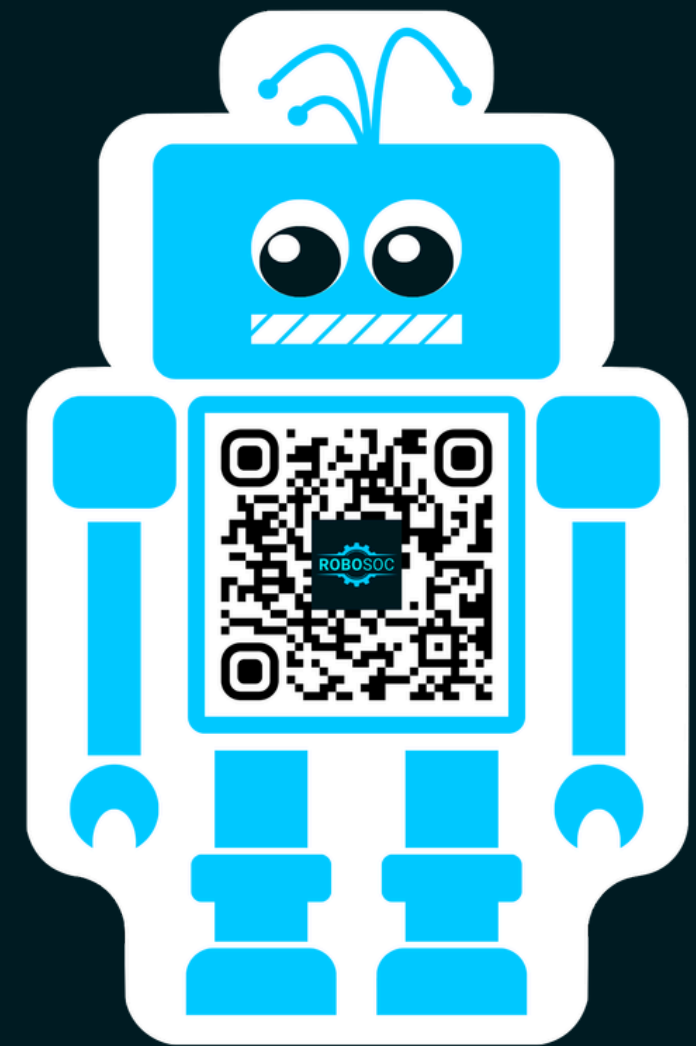
Why do we compete?

- Apply what you learn 
- Build Teamwork skills 
- Develop problem solving skills 
- Meet like-minded people 
- Build your CV 
- Represent Nottingham 
- Have fun 





Unibots



UniBots Competition

- Universities from across the country 
- Design, build and test 
- Travel to different universities 
- Our first year : 7th out of 36 teams 



Unibots Objective

“Build an autonomous robot that can complete the competition challenge and score as many points as possible.”



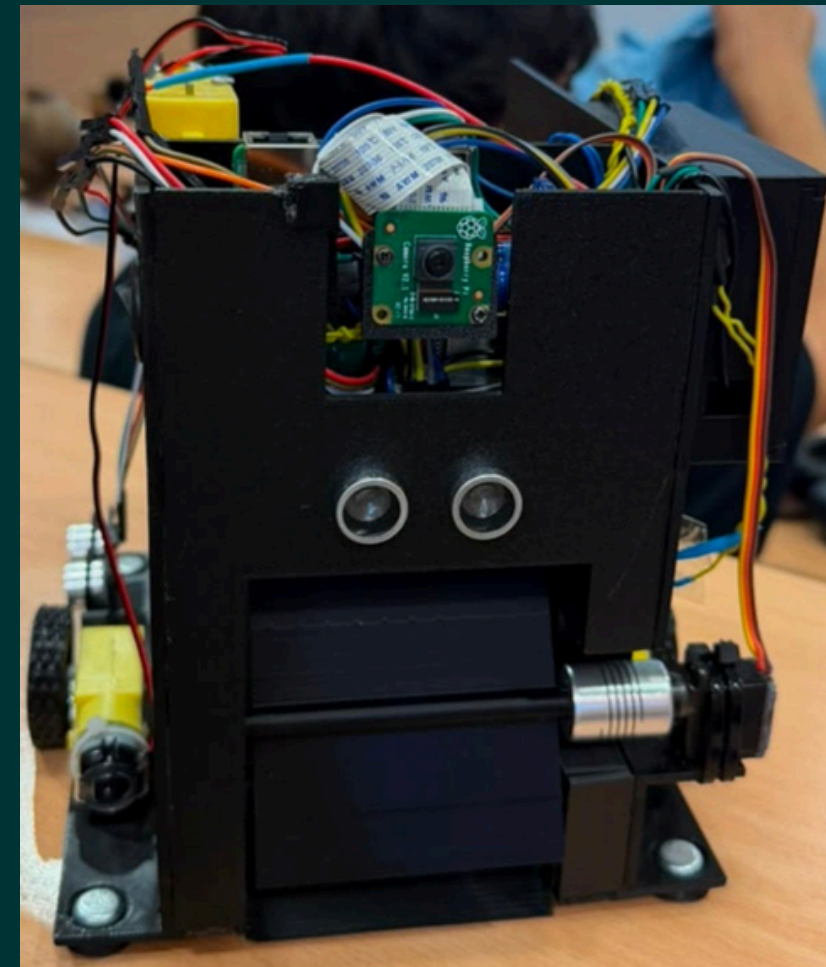
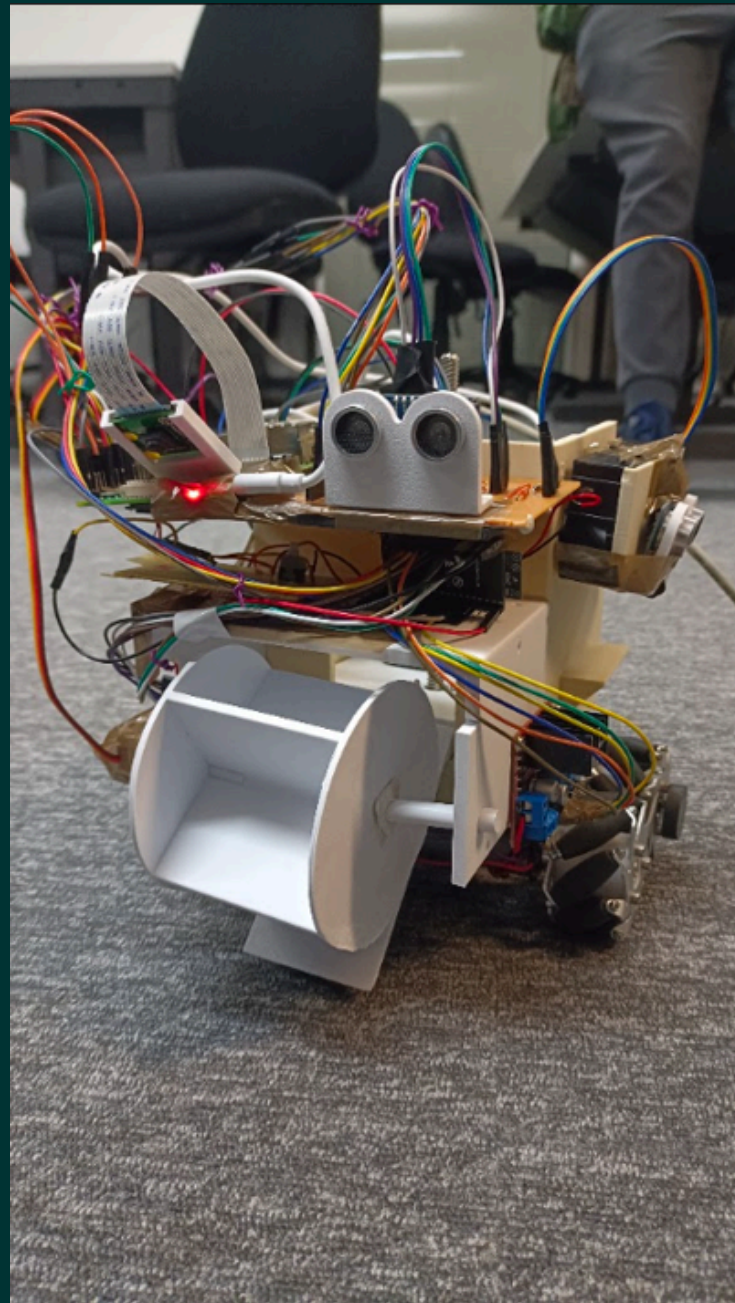
Last Year

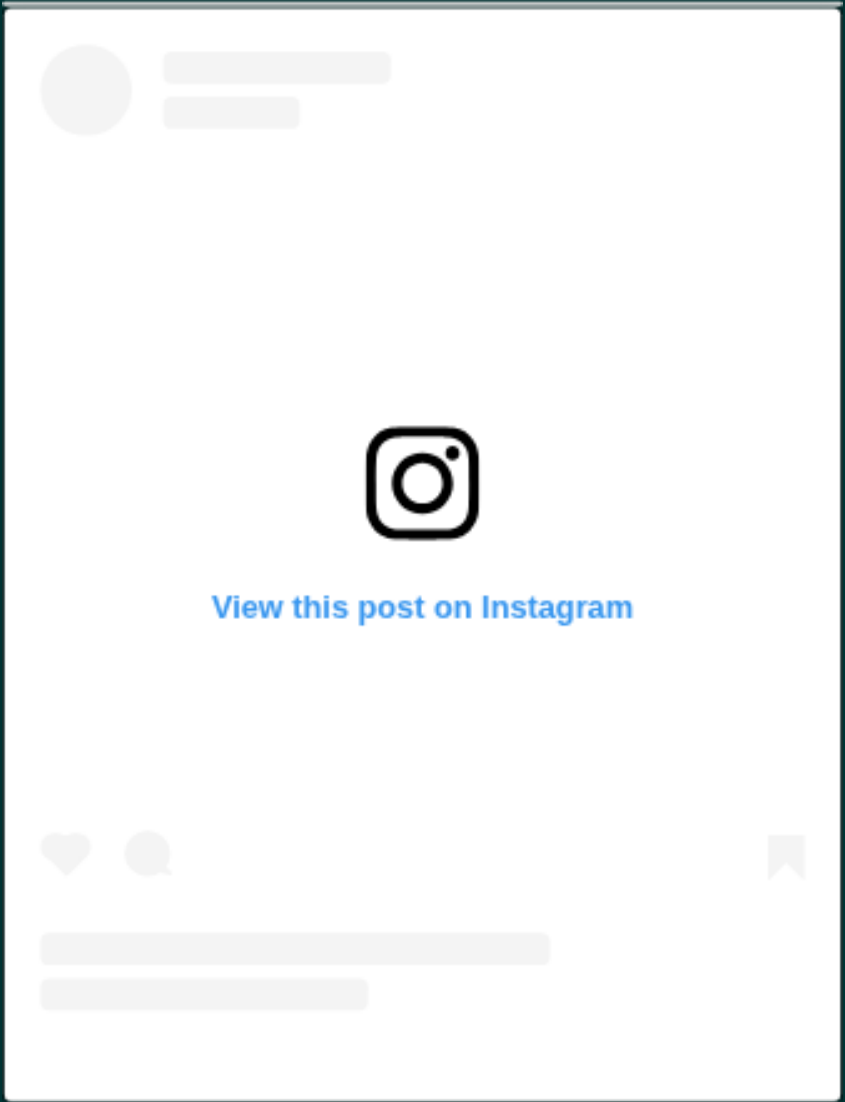
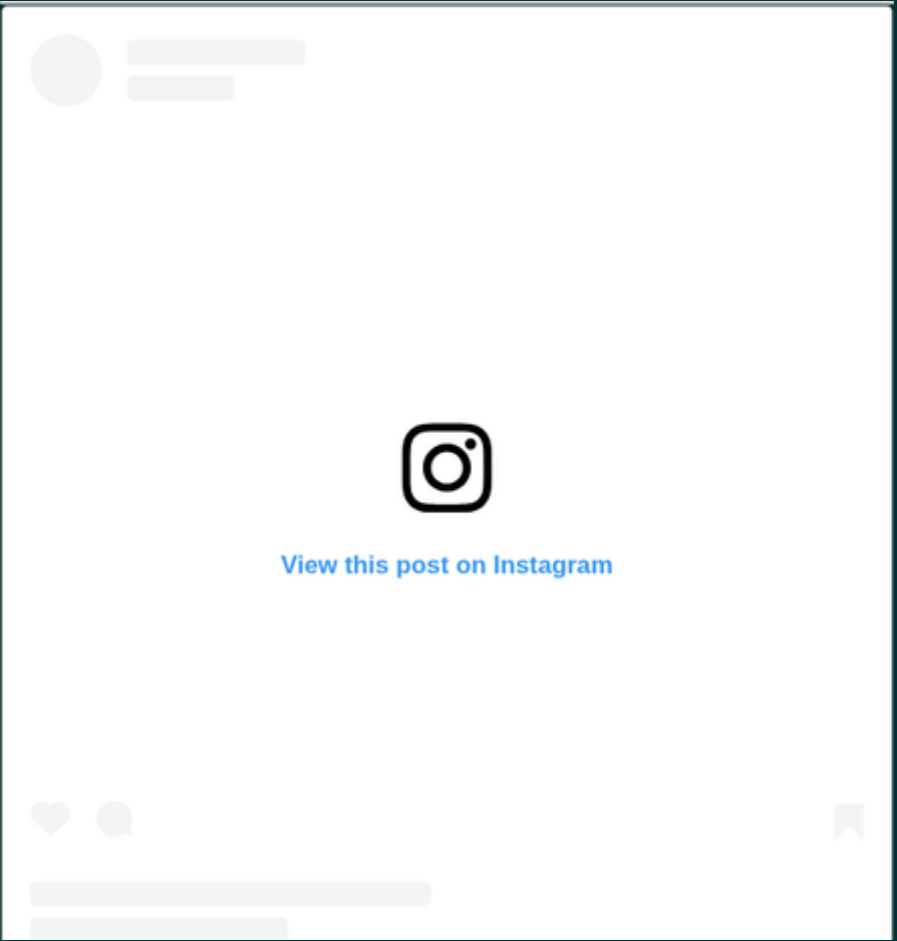
Tasks

- Assemble main chassis
- Parking code
- Re-design Load screw motor holder ✓
- Nameplate (50x50) ✓
- Re-design scroller portion
- Wood Panels???

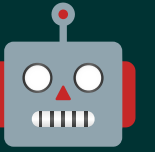
To Do:

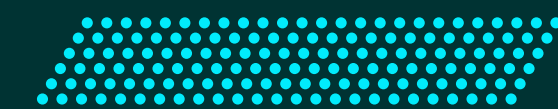
- Print Camera Holder
- Print Name plate
- Print New Scroller Design
- Print New Bottom Motor Holder Design
- Print New Front Chassis Design
- File down/RePrint Back Ultrasonic Sensor
- Length over 230mm(243mm)





This Year

- LEARN -  Workshops covering electronics, programming, sensors, motors & AI
- BUILD -  Develop and improve our competition robot
- TEST -  Regular testing, challenges and practice runs
- COMPETE -  Trials → Final



This Year



Competitive Team (10 members)

- Compete in official UniBots competitions
- Regular building, testing & preparation
- 3–5 hours per week expected



Casual Team

- More flexible involvement
- Take part in builds, testing & internal competitions
- No fixed weekly commitment



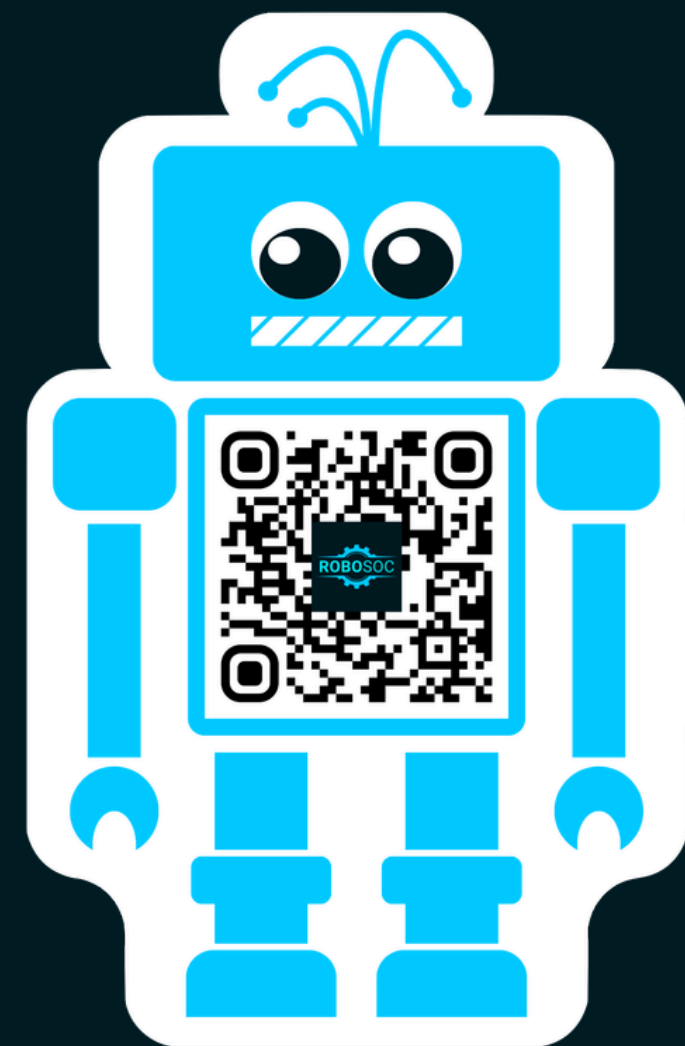
Who do we need?

-  Mechanical - Robot structure, mechanisms & construction
-  Programming - Robot control, algorithms & software, object detection + navigation
-  Electronics - Circuits, wiring, sensors & motors

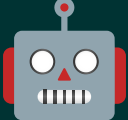




World Robot Olympiad



WRO – World Robot Olympiad

-  Future Engineers & Future Innovators
-  2–3 members per team
-  International robotics competition
-  Design, build & program your robot
-  More details announced soon



WRO – World Robot Olympiad

Future Engineers

- Build and program an autonomous vehicle
- Robot must navigate a competition field without manual control
- Focus on engineering, control systems, sensors and programming
- Teams compete in multiple rounds/challenges
-  Compete against teams from around the world



WRO — World Robot Olympiad

Future Innovators

- Develop a robotics solution to a real-world problem
- Design, build and program your own robot/system
- Focus on innovation, creativity and problem-solving
- Present your project and explain how your solution works
-  Compete against teams internationally



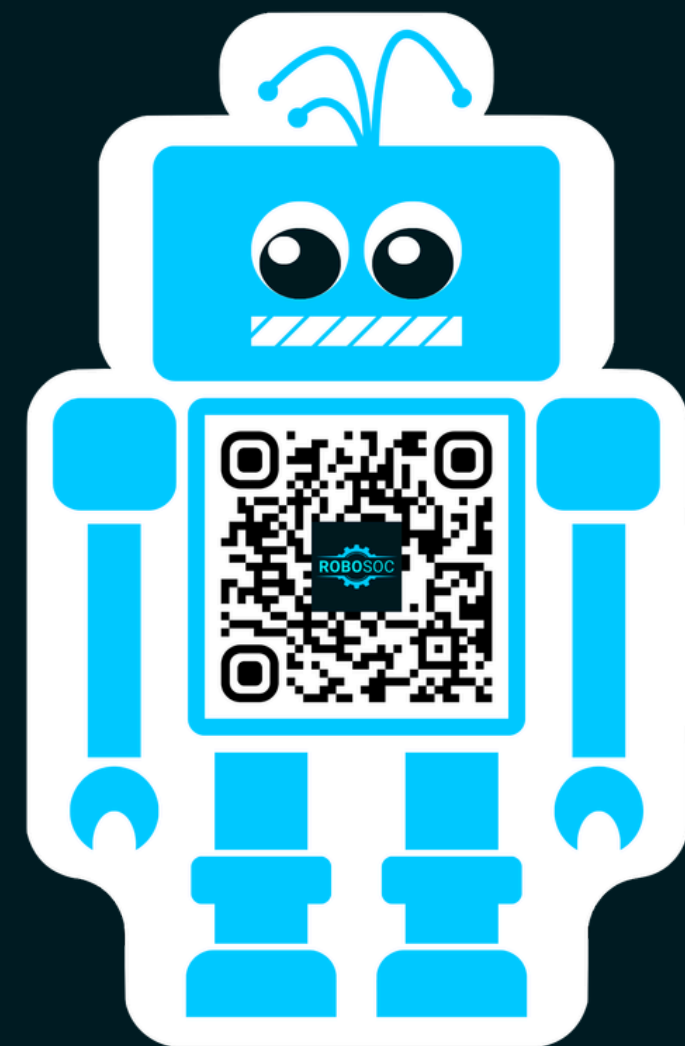
You don't need to be an expert!

We're looking for people who are willing to learn
and contribute!

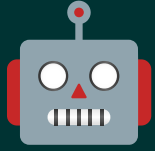


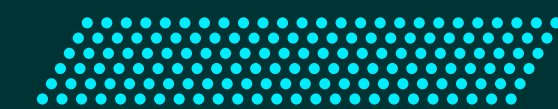


Internal Competition

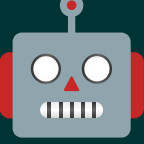


Internal Competition

-  Lower-pressure competition – less intense than official competitions
-  On campus – no travel required
-  Free to enter – open to everyone
-  Similar to the EEE First Year Robot Car Project – build and compete with your robot



READY TO COMPETE?

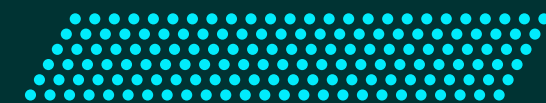
-  Come to the workshops
-  Join the competition team
-  Become a member
-  Talk to us after the session
-  Scan the QR code / join Discord



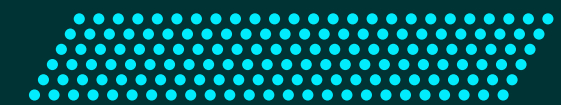
Volleyball IMS

-  Represent Robotics Society in University IMS volleyball
-  Play with other society members
-  No experience needed - just come along and have fun!
-  Compete against other university teams
-  Details & sign-up coming end of the week





Keychains



Join ROBOSOC TODAY!



Competition
Team
Sign-Up



Discord



LinkTree
(ALL our links)



Instagram



IMS Team
Sign-Up

